Alpha decay 3

In this section we derive C_3 from the following half-life data. Rows with $Z = 90$ are uranium and $Z = 88$ are thorium. Recall that Z is atomic number minus two.

$\log \tau_{1/2}$	A	Z	E
6.30	228	90	6.80
14.4	230	90	5.99
21.5	232	90	5.41
29.7	234	90	4.86
34.2	236	90	4.57
39.5	238	90	4.27
7.52	226	88	6.45
17.9	228	88	5.52
28.5	230	88	4.77
40.6	232	88	4.08

From the previous section we have

$$\log \tau_{1/2} = 2\left(C_1 \frac{Z}{\sqrt{E}} - C_2\sqrt{Zr_1}\right) + C_3 \log A + \log(\log 2)$$

Hence the regression model

$$\log \tau_{1/2} - 2\left(C_1 \frac{Z}{\sqrt{E}} - C_2\sqrt{Zr_1}\right) - \log(\log 2) = C_3 \log A$$

By ordinary least squares we obtain

$$C_3 = -10.1$$

The regression formula yields the following predicted values.

$\log \tau_{1/2}$	Predicted $\log \tau_{1/2}$
6.30	5.31
14.4	14.0
21.5	21.4
29.7	29.7
34.2	34.5
39.5	40.1
7.52	6.87
17.9	17.8
28.5	28.8
40.6	41.6

Eigenmath script